Value Added Problems

Problem 1: Print all values in the BST within a given range Approach:

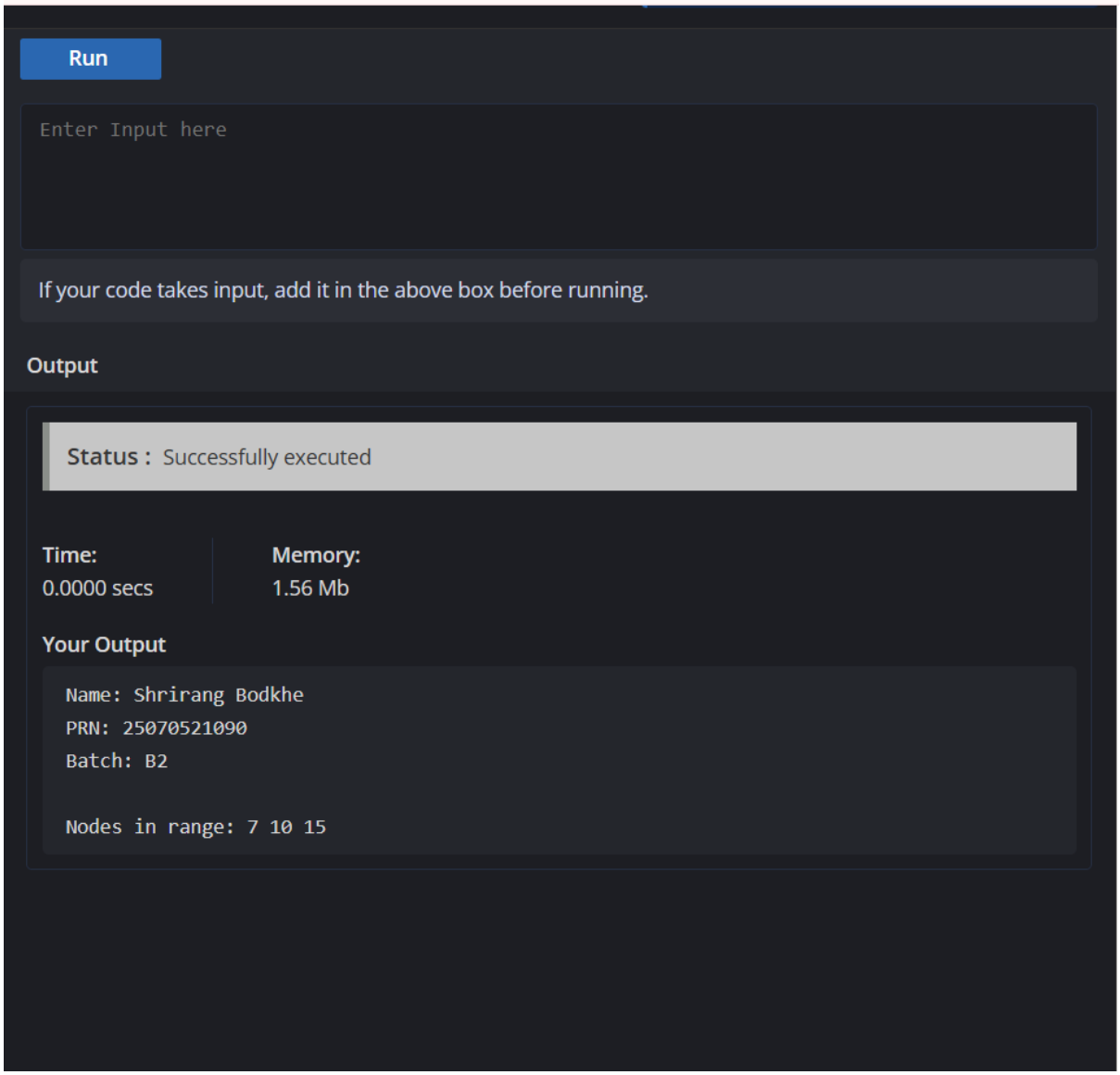
Use DFS traversal (in-order) and prune branches using BST properties:

- If node value is greater than low, explore left subtree
- If node value is within range, include it
- If node value is less than high, explore right subtree

Problem 1: BST Range Query #include <stdio.h>

```
#include <stdio.h>
#include <stdlib.h>

struct Node {
    int data;
    struct Node * left;
    struct Node * right;
};
struct Node * createNode(int val) {
    struct Node * node = (struct Node * ) malloc(sizeof(struct Node));
    node -> data = val;
    node -> left = node -> right = NULL;
    return node;
}
void rangeQuery(struct Node * root, int low, int high) {
    if (root == NULL) return;
    if (root -> data > low)
        rangeQuery(root -> left, low, high);
    if (root -> data >= low && root -> data <= high)
```

```
        printf("%d ", root -> data);
    if (root -> data < high)
        rangeQuery(root -> right, low, high);
}
int main() {
    printf("Name: Shrirang Bodkhe\n");
    printf("PRN: 25070521090\n");
    printf("Batch: B2\n\n");
    struct Node * root = createNode(10);
    root -> left = createNode(5);
    root -> right = createNode(15);
    root -> left -> left = createNode(3);
    root -> left -> right = createNode(7);
    root -> right -> right = createNode(18);

int low = 7, high = 15;

 printf("Nodes in range: ");
 rangeQuery(root, low, high);
    printf("\n");

return 0;
}
```



Problem 2: Check if expression is balanced Approach:

Use a stack:

- **Push opening brackets**
- **On closing bracket, check top of stack**
- **If mismatch or empty → not balanced**
- **At end, stack must be empty**

Problem 2: Balanced Brackets (C)

```
#include <stdio.h>
#include <string.h>

#define MAX 1000000

char stack[MAX];
int top = -1;
void push(char c) {
   stack[++top] = c;
}
char pop() {
   return stack[top--];
}
int isEmpty() {
   return top == -1;
}
int isBalanced(char * s) {
   for (int i = 0; s[i] != '\0'; i++) {
      char c = s[i];
      if (c == '(' || c == '{' || c == '['
         push(c);
      else {
         if (isEmpty()) return 0;
         char t = pop();
         if ((c == ')' && t != '(' ||
            (c == '}' && t != '{') ||
            (c == ']' && t != '[')
```

```
 return 0;
        }
    }
    return isEmpty();
}
int main() {
    char s[] = "{[()]}";
    printf("Name: Shrirang Bodkhe\n");
    printf("PRN: 25070521090\n");
    printf("Batch: B2\n\n");
    if (isBalanced(s))
        printf("Balanced\n");
    else
        printf("Not Balanced\n");
    return 0;
}
```